Social Media Engagement Analytics - One-Page Summary

Dataset: 5,000 posts, 10 countries, Jan 2022 - Dec 2023 | Tools: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy

Data preparation

150 missing values in each of six columns were imputed (median/mode; median by post type for likes, comments, shares; "unknown" kept for sentiment). No fully duplicated rows. 467 rows with likes, comments or shares above impressions were flagged and capped. Engagement rate is extremely skewed (median 0.25, mean 0.96, max 191), so rankings use the **median**.

Headline finding

No group difference is statistically significant (all p-values > 0.05). Post type, category, country, age, device and sentiment all differ by only a few percent. The "best" groups below are numerically highest, not proven drivers of engagement. The metrics look uniformly random and uncorrelated, which suggests synthetic data.

Key results

Question	Answer	Support
Best post type	Reel (11,981 interactions/post); image a near tie and top median rate (0.26)	Spread < 1%, p = 0.69
Best category	Music (12,133 interactions); education has top median rate (0.28)	p = 0.98
Top countries (median rate)	UAE, Australia, Germany tie at 0.27; USA, UK, France lowest at 0.24	p = 0.36
Age effect	None: corr with rate 0.01; 18-24 median 0.27 vs 55-64 0.24	p = 0.54
Verified accounts	No advantage: 11,518 vs 11,988 interactions; median rate 0.24 vs 0.26	p = 0.09
Best time of day	Not measurable: dates only, no clock time. Proxy: Tuesday highest impressions (51.3k), Thursday lowest (48.1k)	p = 0.46
Device vs watch time	Mobile highest (4,088 s) vs desktop 3,975 s, tablet 3,980 s	p = 0.28
Best sentiment	Positive leads on interactions (12,003) and is 48.7% of posts; neutral is lowest (11,793)	p = 0.99
Negative / neutral posts	Negative (19.8%) has the most shares (1,001) and highest mean rate (1.04); neutral (31.5%) travels least	not significant

Recommendations

1. Do not change content strategy on these small gaps; run controlled A/B tests by post type first.
2. Record hour-level posting time so best-time-of-day can be analysed.
3. Investigate the 68 low-impression posts with engagement rate above 10, which are the true outliers.
4. Report median (or capped) engagement rate, never the raw mean.